

CHE374 — Problem Set 1 Solutions

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Problem Set 1

Question 1

A new steel stirred tank reactor costs \$8,000, plus \$1,200 for installation. The company has an old stainless steel unit in the warehouse that could be used instead. The old reactor was used for 4 years but will last the same length of time as the new one; it cost \$16,000 when purchased. Installing the old reactor would cost \$2,500. The old reactor has a used market value of \$6,800.

a) Determine the best alternative.

b) What market value would make you indifferent between the two?

Solution: The \$16,000 original purchase price of the old reactor is a **sunk cost** — already spent, and irrelevant to the decision going forward. What matters is the *opportunity cost* of using the old reactor instead of selling it at its market value. Since both reactors have the same remaining life, we can compare total up-front cost directly.

- New reactor: $8,000 + 1,200 = \$9,000$.
- Old reactor: $6,800 + 2,500 = \$9,300$ (market value forgone, plus installation).

a) Since $\$9,000 < \$9,300$, buy the **new reactor**.

b) Let X be the old reactor's market value. Indifference means

$$X + 2,500 = 9,000 \implies X = \$6,500.$$

Since the actual market value (\$6,800) is above this breakeven point, the old reactor is relatively too valuable to give up for free — confirming the new reactor is the better buy.

Question 2

A restaurant is upgrading its cash registers. Company X quotes \$7,500 up front, saving \$1,700/year. Company Y quotes \$9,000 up front, saving \$2,200/year.

- a) *How long to recover the initial investment for each?*
 b) *List at least one issue with this type of comparison.*

Solution:

a) Payback period = initial cost / annual savings.

$$\text{Company X: } \frac{7,500}{1,700} \approx 4.41 \text{ years,} \quad \text{Company Y: } \frac{9,000}{2,200} \approx 4.09 \text{ years.}$$

Company Y pays back faster despite the larger upfront cost.

b) Simple payback analysis has (at least) two issues:

- It ignores the **time value of money** — future savings are not discounted back to present worth.
- It ignores everything that happens *after* the payback period (e.g. total savings over the equipment's full life, or differences in equipment lifespan/reliability between options).

Question 3

$$\text{Find } \lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^n.$$

Solution: This is exactly the continuous-compounding limit from the notes (with the “−1” dropped):

$$\lim_{n \rightarrow \infty} \left(1 + \frac{r}{n}\right)^n = e^r.$$

Question 4

An 8-year \$100,000 loan charges an effective 3.4% interest per year. Find the nominal annual rate for: a) semi-annual, b) monthly, c) daily, d) continuous compounding. e) How much is owed after 4 years in each case?

Solution: Solve $i_e = (1 + \frac{r}{m})^m - 1 = 0.034$ for r , i.e. $r = m \left[(1.034)^{1/m} - 1 \right]$.

- a) $m = 2$: $r = 2 \left[(1.034)^{1/2} - 1 \right] \approx 3.372\%$.
- b) $m = 12$: $r = 12 \left[(1.034)^{1/12} - 1 \right] \approx 3.348\%$.
- c) $m = 365$: $r = 365 \left[(1.034)^{1/365} - 1 \right] \approx 3.343\%$.
- d) Continuous: $r = \ln(1.034) \approx 3.343\%$ (matches $e^r - 1 = i_e$ from the notes).

e) Since every case was solved to give the *same* effective annual rate (3.4%), the amount owed after 4 years is identical regardless of compounding frequency:

$$F_4 = 100,000(1.034)^4 \approx \$114,309.46 \quad \text{in all four cases.}$$

This is the point of the effective rate: it's the true, comparable growth rate, independent of how compounding is described.

Question 5

A \$100 T-bill (face value paid at maturity, no coupons) sells at a continuously compounding yield of 2.3%/year.

a) Effective annual yield with yearly compounding?

b) Price if it matures in 6 months? c) In 9 months?

d) If the yield drops 30 bps (to 2%), how do the prices in b) and c) change?

Solution: For continuous compounding, price = Fe^{-rt} where $F = \$100$ is the face value and t is time to maturity in years.

a) $i_e = e^{0.023} - 1 \approx 2.327\%$.

b) $t = 0.5$: Price = $100 e^{-0.023(0.5)} \approx \98.86 .

c) $t = 0.75$: Price = $100 e^{-0.023(0.75)} \approx \98.29 .

d) With $r = 2\%$:

6-month: $100 e^{-0.02(0.5)} \approx \99.00 (+\$0.15), 9-month: $100 e^{-0.02(0.75)} \approx \98.51 (+\$0.22).

Both prices *increase* when the yield drops — price and yield move in opposite directions for a zero-coupon bond — and the 9-month (longer maturity) price is more sensitive to the yield change than the 6-month price.

Question 6

Interest rate conversion table (assume 252 trading days/year where a daily rate is involved).

Solution:

1. **Given** 8% effective/year $\rightarrow$ **monthly rate, monthly compounding:** $i_s = (1.08)^{1/12} - 1 \approx 0.643\%$ /month.
2. **Given** 3.5%/year nominal, daily compounding ($m = 252$) $\rightarrow$ **effective rate/year:** $i_e = \left(1 + \frac{0.035}{252}\right)^{252} - 1 \approx 3.562\%$.
3. **Given** 4%/year nominal, quarterly compounding ($m = 4$) $\rightarrow$ **continuous rate/year:** $r_c = 4 \ln(1.01) \approx 3.980\%$.
4. **Given** 1.5%/month, monthly compounding $\rightarrow$ **continuous rate/quarter:** 3 months per quarter, so $r_c = 3 \ln(1.015) \approx 4.467\%$ /quarter.
5. **Given** 1.2%/quarter, monthly compounding ($m = 3$ months/quarter, so 0.4%/month) $\rightarrow$ **interest per 3 years, semi-annual compounding:** the equivalent effective rate is $(1.004)^{36} - 1 \approx 15.455\%$ total over 3 years (equivalently, an effective semi-annual rate of $(1.004)^6 - 1 \approx 2.423\%$, compounded over the 6 half-years in 3 years).